Promit Moitra

Data & ML Engineer · Computational Neuroscience & Complex Systems

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Full-stack data and ML engineer with a physics PhD in complex systems. I build the backends and pipelines that turn raw multi-sensor recordings into **ML-ready datasets and evidence-backed results** — with a bias toward leakage-aware evaluation, honest negatives, and audit trails you can verify. Background spans computational neuroscience, nonlinear dynamics and statistical signal processing.

EXPERIENCE

R&D Scientist — Data & ML Engineering · *Dognosis India Pvt. Ltd, Bangalore* Jun 2025 – Present

- Sole-architected `resp` (984-commit R&D pipeline): GOR sniff detection, LSL↔Unix clock synchronization, three-way sample attribution, a logical-session resolver, and a versioned star-schema analytics layer (MotherDuck).
- Top contributor to `dogapi`, the production FastAPI backend, and sole author of the dataset-generation and session-validation vertical — fusing five evidence streams (IR array, wireless LSL, vision, QR, trainer-app state) into one confidence-tiered record per sniff behind a physiological validation gate.
- Designed and built `otvcore`: a git-backed, tamper-evident design-control system of record (OTV / NCR / FMEA / ECO / SOP) as one engine with three clients (FastAPI + React UI, CLI, agent skill), gated by a CI verify step.
- Data-integrity forensics: root-caused silent training-pipeline corruption — crash-restart session contamination, a ~57%-duration capture, and a ~30% sample-code mismatch from a QR-scan/TimescaleDB timing race — each with per-timestamp evidence and a precise fix.
- Canine EEG & olfaction stack (SniffSpace, DogSense): multi-sensor acquisition (EEG, respiration, IMU, video), Lab Streaming Layer synchronization, signal QC, and CNN classification of neural responses.

Postdoctoral Researcher · *Decision Lab, IIT Kanpur* Oct 2022 – May 2025

- Studied neural correlates of stress in a patch-foraging paradigm — EEG with behavioral analysis and convolutional neural networks for response classification.
- Awarded a two-year **DST Cognitive Science Research Initiative (CSRI) Post-Doctoral Fellowship** for “What drives patch foraging behavior: a dynamical-systems account of neurocognitive mechanisms.”
- Contributed to the **Rice–IITK Strategic Collaboration** on high-density, multi-region intracortical depth-array mapping during decision tasks — presented by invitation at **InterfaceRice 2024** (Rice University, Houston) and co-authored the resulting *NeuroImage* (2026) paper.

Postdoctoral Fellow · *Institute for Plasma Research, Gandhinagar — Basic Theory & Simulation Division* Nov 2019 – Sep 2022

- Modeled excitable media with cellular automata, showing how local timescales shape spatiotemporal dynamics (published in *Chaos, Solitons & Fractals*, 2022).

PUBLICATIONS

J. A. Willis, C. E. Wright, …, **P. Moitra**, …, J. P. Seymour (2026). Optimizing electrode placement and information capacity for local field potentials in cortex. *NeuroImage*, 327, 121747. doi:10.1016/j.neuroimage.2026.121747

P. Moitra, A. Sen (2022). Impact of local timescales in a cellular automata model of excitable media. *Chaos, Solitons & Fractals*, 162, 112418. doi:10.1016/j.chaos.2022.112418

P. Moitra, S. Sinha (2019). Localized spatial distribution of disease phases yields long-term persistence of infection. *Scientific Reports*, 9, 20309. doi:10.1038/s41598-019-56616-3

P. Moitra, S. Sinha (2019). Emergence of extreme events in parametrically coupled chaotic oscillators. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 29, 023131. doi:10.1063/1.5063926

P. Moitra, K. Jain, S. Sinha (2018). Anticipating persistent infection. *Europhysics Letters*, 121, 60001. doi:10.1209/0295-5075/121/60001

V. Agrawal, **P. Moitra**, S. Sinha (2017). Emergence of persistent infection due to heterogeneity. *Scientific Reports*, 7, 41582. doi:10.1038/srep41582

EDUCATION

Integrated MS–PhD, Physics · *IISER Mohali* 2012 – 2019

- Thesis: *Dynamics on Spatially Extended Systems* — nonlinear dynamics, chaos, complex networks, and models of disease spread. Supervisor: Prof. Sudeshna Sinha.

BSc, Physics (First Class) · *Loyola College, Chennai* 2009 – 2012

AWARDS & DISTINCTIONS

- **DST CSRI Post-Doctoral Fellowship** — two-year fellowship, Cognitive Science Research Initiative, Dept. of Science & Technology, Govt. of India.
- **Indian Academy of Sciences Summer Fellowship** (2013) — Institute for Plasma Research, Gandhinagar.
- **IIT-JAM 2012** — All India Rank 477 (~50,000 applicants); admission to the Integrated MS–PhD program at IISER Mohali.

SELECTED TALKS, SCHOOLS & VISITS

- **InterfaceRice 2024**, Rice University, Houston — invited participant, Rice–IITK collaboration.

- **Brain, Computation and Learning** (2023), Indian Institute of Science, Bangalore.
- **School & Workshop on Patterns of Synchrony** (2019), ICTP, Trieste — poster, "Anticipating Persistent Infection."
- **Research visit** (2017), Potsdam Institute for Climate Impact Research (PIK) — networks of coupled maps, w/ Prof. Jürgen Kurths.
- **PHYSCON 2017**, University of Florence — talk, "Heterogeneity Facilitates Persistent Infection."
- **Santa Fe Complex Systems Winter School** (2015), IISER Mohali.

SKILLS

Backend & Data · Python, SQL, FastAPI, Pydantic v2, async SQLAlchemy/asyncpg, PostgreSQL, TimescaleDB, DuckDB/MotherDuck, Docker, GitLab CI, TypeScript, React

Modeling & ML · Convolutional & sequence models, leakage-aware cross-validation, nonlinear dynamics & chaos, complex networks (NetworkX), scikit-learn, Pandas, NumPy

Signals & Neuro · EEG, respiration & IMU processing, Lab Streaming Layer, multi-sensor fusion & clock sync, SciPy, MNE-Python, signal quality control

Also · MATLAB, C++, Git, HPC job scheduling (PBS), Bash, LaTeX; Windows & UNIX

PEER REVIEW & SCIENCE COMMUNICATION

- Reviewer, *AIP Chaos* (2022).
- Moderator & interviewer, India Science Festival (2021–2022) — including "Emerging Paths of Quantum Computing" with Ilyas Khan.

INTERESTS & LANGUAGES

- **Languages:** English, Hindi, Bengali (fluent — speaking, reading, writing); French (basic reading & writing).
- **Beyond work:** 15+ years of Indian classical music; lead vocals in a college rock band; national-level parliamentary debate (convenor, IISER Mohali debating society); football (school-team captain).

REFEREES

Prof. Sudeshna Sinha
Deputy Director & Professor, Dept. of Physical Sciences, IISER Mohali
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Prof. Abhijit Sen
Emeritus Professor & INSA Sr. Scientist, Institute for Plasma Research
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Prof. Subroto Mukherjee
Dean (Administration) & Professor, Institute for Plasma Research
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